

William Rooney

(843)-441-7389 · wcroone@clemson.edu

· www.linkedin.com/in/william-rooney-ba565a3aa

OBJECTIVE

To earn a design-focused motorsports internship where I can contribute to race-ready powertrain/driveline hardware through CAD, analysis, rapid prototyping, and data-driven validation.

EDUCATION

Bachelor of Science in Automotive Engineering

Clemson University

Emphasis: Powertrain Performance and Development, Vehicle Dynamics

Minor: Mathematical Sciences

Spring 2028
Clemson, SC

RELEVANT EXPERIENCE

Lead Drivetrain Design Engineer

Formula SAE, Clemson University

Summer 2025 - Present
Clemson, SC

- Drive cross-team interface resolution (chassis, suspension, powertrain) by managing packaging/clearance constraints and communicating decisions in-person and via Slack
- Designed and released drivetrain components in SolidWorks; verified strength/stiffness in ANSYS; iterated from test feedback to improve strength, packaging clearance, and reduce weight
- Implement the new design into the new season car and make quick, decisive actions regarding the drivetrain system throughout the season when issues arise

Powertrain Division Member

Formula SAE, Clemson University

Fall 2024 - Present
Clemson, SC

- Contribute to troubleshooting plans using observed symptoms and measurements; document fixes and prevention steps across competition cycles
- Support engine and drivetrain reliability through hands-on diagnostics, teardown, rebuild, and reassembly between events and competitions

Full Vehicle Performance Restoration & Modification Project

Personal Passion Project

Winter 2022 - Present
Beaufort, SC

- Re-engineered and assembled a fully operational Volkswagen GTI for track/street performance by sourcing, inspecting, and refurbishing salvaged components from sibling models with better performing components, creating a cost-efficient vehicle
- Implemented sustainable engineering practices by sourcing and rebuilding an engine from a donor vehicle, extending life-cycle and reducing build costs
- Diagnosed and repaired electrical, powertrain, and suspension systems when issues arose; verified repairs through functional checks and road testing

WORK EXPERIENCE

Sales Associate

Advance Auto Parts

Winter 2024 - Summer 2024
Beaufort, SC

- Utilized AdvancePro and Z-net systems to diagnose vehicle issues and identify precise automotive parts
- Conducted in-store diagnostic services, including battery, alternator, and starter testing, to identify component failures for customers

Certifications

Certified SolidWorks Associate

Lean Six Sigma Green Belt

Spring 2025
Spring 2026

TECHNICAL SKILLS

SolidWorks - Proficient

MATLAB - Proficient

ANSYS Mechanical - Basic

Microsoft Excel - Proficient

